

Cooperation by Default: LLM Agents in the Collective-Risk Dilemma

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Abstract

Language-model agents now act for people in shared settings, where one agent’s choice can put a whole group at risk. We ask whether such agents cooperate because they weigh the risk and their partners, or by default, with play that stays the same whatever is at stake. In the collective-risk social dilemma, players pay into a shared pool over several rounds, and if the pool misses a target, a catastrophe may wipe out everyone’s savings. A risk-neutral player should keep its money when the risk is low and pay a fair share when it is high. We test five low-cost models (Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20) in self-play across risk levels, under prompt changes, beside scripted partners and at tables that mix two models. Most models pay about the same amount at every risk level, even when there is no risk at all. Removing moral and climate words changes nothing. Telling the agent that only its own money matters stops most models from paying at zero risk, but only one then pays little at low risk and a fair share at high risk. Several models say correctly which choice pays more, yet their payments do not change, and no model keeps all its money beside partners who make every payment a loss. In mixed groups, one agent that stops paying at the end breaks a table of exact fair-share players. Under selection by payoff, comparing agents with their tablemates rewards the smallest payer, while comparing them across groups keeps the fair-share player when the risk is not low. Agent evaluations should state the agent’s goal, compare play with the optimum and use mixed populations.

Keywords

LLM agents; collective-risk social dilemma; cooperation; best response; empirical game-theoretic analysis; multiagent evaluation

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1 Introduction

Language-model agents are starting to act for people in shared settings: they spend budgets, share compute and bargain over common resources. Each such agent is a player in a social dilemma, because what it pays helps the group and costs its owner. Before deploying

one, a designer needs to know two things. Does the agent’s cooperation depend on what is at stake? And does it depend on what the other agents do?

The most common test, self-play, cannot answer either question. When every seat holds a copy of the same model, an agent that reasons about the stakes and an agent that repeats a fixed rule can leave the same record of play. Many studies also report whether the group reached its goal, not how far the agents were from the best they could have done, yet a group can reach its goal and still waste most of its money. And a prompt that gives a group target and a number of rounds already suggests an equal split, so an agent that simply repeats that split looks cooperative on every summary measure.

We study both questions in a classic game from research on climate cooperation, the collective-risk social dilemma [17]. Players pay into a shared pool over several rounds, and if the pool misses a target, a catastrophe destroys everyone’s savings with probability p . Rational play in this game has a simple shape. For risk-neutral players, nobody should pay while the risk is below one half, and above it everyone should pay the *fair share*, the target split equally. When there is no risk at all, paying only lowers a player’s own money, so no attitude to risk can explain it.

We call play a *default* when it does not change with what is at stake. The game offers two sharp tests of a default, because in both situations paying cannot help: paying when there is no risk, and paying after the outcome can no longer change. We test five low-cost production models, one per provider: Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20. We use four designs. Self-play covers the whole range of risk. Prompt controls ask the agent which choice pays more, remove normative words, or state the agent’s goal. One model plays beside scripted partners, where the best response can be computed exactly. And two different models share a table, which lets us ask which agent would win if agents were selected by payoff [29, 31]. We report four findings (Figure 1).

- **Risk does not change the default** (Section 4). Most models pay about the fair share at every positive risk level, and most also pay when there is no risk. Removing normative words changes neither result. Stating that only the agent’s own cash matters stops most models from paying at zero risk, but only GPT-5.6 Luna then pays little at low risk and the fair share at high risk.
- **Answers track the risk; play does not** (Section 5). Asked in separate calls, several models rank the two choices correctly and change that ranking with the risk, but they keep paying the same amount. Beside scripted partners who make every payment a loss, no model keeps all its money, and two models keep paying after the outcome is settled.



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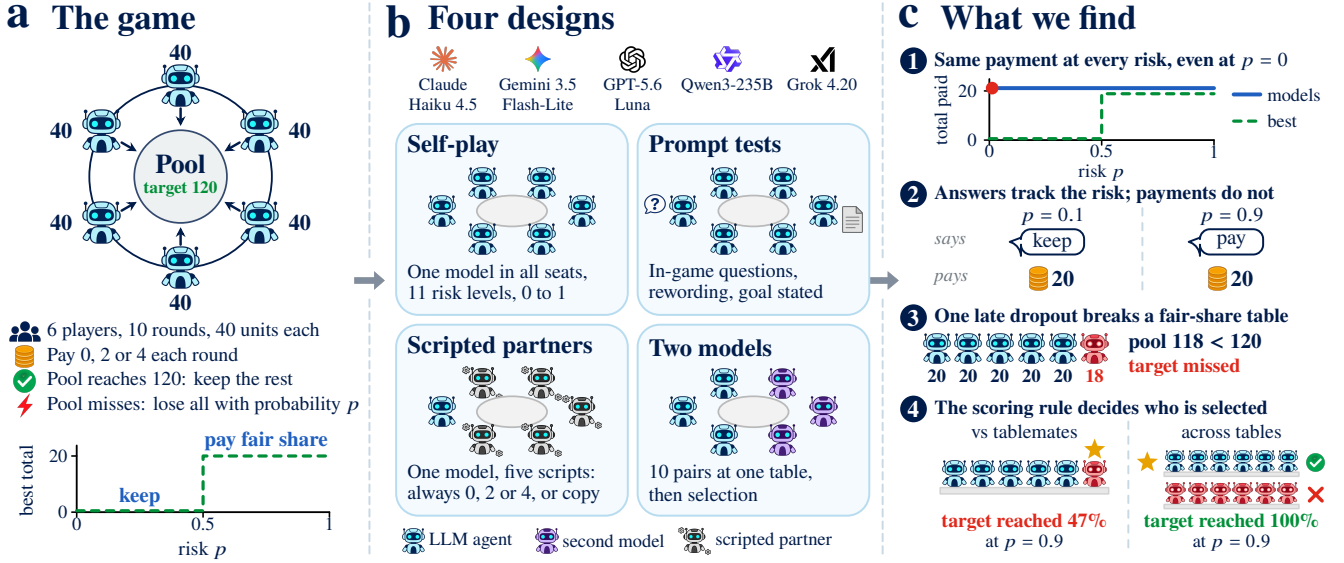


Figure 1: Overview. (a) Six players hold 40 units and pay 0, 2 or 4 into a shared pool in each of ten rounds; if the pool misses 120, all savings are lost with probability p . Below: the best total per seat (Proposition 1). (b) The five models and the four designs, with who sits at each table. (c) The four findings, in the order of Sections 4 to 7; the target rates in the last are those of the population that selection favours at $p = 0.9$ (Figure 7).

- **An exact fair share has no slack** (Section 6). One partner that stops paying in the last round breaks a table of exact fair-share players, while a table that overpays a little absorbs it. Agents pay less when generous partners fill most of the table.
- **The scoring rule decides who is selected** (Section 7). Seats at one table share the pool and the lottery, so the smaller payer always earns more there. Scoring agents against their tablemates therefore favours the smallest payer and, in our games, lowers group success. Scoring them across separate groups favours the exact fair-share player when the risk is moderate or high.

For the models we test, cooperation is a default that follows from what the prompt leaves unsaid, not a choice that follows from the stakes. A default can work well beside partners like itself, but it cannot be trusted with shared resources, and the way a system states its agents' goals and selects its agents can make the problem better or worse.

2 Related work

Collective risk and selection. The collective-risk dilemma was built to study why groups fail to avoid a shared disaster [17]. Human groups reach the target mostly when the risk is high, and inequality and communication change the result [27]. Games with a threshold have many equilibria [19], human contributions decline over the rounds of repeated public goods games [2], and the risk of collective failure can turn a defection trap into a coordination problem [22]. Theory also shows that selection on payoff relative to a small group favours whoever does better than its neighbours [23], while selection between groups can favour cooperation [28]. This

literature tells us that behaviour *should* move with the risk, so a flat response is informative, and we reuse its selection theory to read our mixed tables.

Language models in games. Language models have been studied as economic subjects [4, 9, 16], in repeated two-player games [1, 10], and in multi-player game benchmarks [12]. Models tend to be more cooperative than people [16], an even split can reflect a focal point rather than fairness [6], and models can hold a correct belief or plan and act against it [7, 24]. In social dilemmas, societies of agents over-use a common resource [21], reasoning models free-ride in public goods games [11], and agents stand in for people in a collective-risk game [25]. Closest to us, Willis et al. study the same collective-risk game with strategies written by three models, map welfare over mixtures of selfish and collective strategies, and find that payoff-biased imitation drives larger groups to selfish, low-welfare outcomes [32]. We differ in three ways. We record the turn-by-turn play of deployed models, not code they write. We compare that play with a computed optimum and add two tests where paying cannot help. And we show that whether selection compares tablemates or separate groups decides which model wins, which gives one simple route to selfish outcomes.

Multiagent evaluation. Sequential social dilemmas made cooperation a property of learned policies [13, 15], and cooperative AI asks for agents that work with partners unlike themselves [5, 26]. Melting Pot tests a focal population against background agents it has never met [14]. Empirical game-theoretic analysis treats trained agents as strategies of a meta-game [29, 31], and α -Rank ranks them by evolutionary dynamics [18]. Tournaments that mix language models with classic strategies report model-specific strategic fingerprints in the prisoner's dilemma [20], and cooperation evolves

Table 1: Symbols and terms.

Term	Meaning
Seat; table	One player position; the six seats of one game
Risk p	Chance that a missed target destroys all savings
Fair share	The target split equally: 20 per seat, or 2 per round
Exact split	Paying 2 in every one of the ten rounds
Pivot $p^* = \frac{1}{2}$	Risk above which reaching the target pays
$\text{opt}(p)$	Best attainable expected payoff per seat
Dominated	Lowers the seat’s own payoff whatever others do
Settled	The history proves that more payment cannot change the outcome: the pool has reached T , or cannot reach it even if every seat pays 4 from now on
Default	Play that does not change with what is at stake

differently across base models in a donor game [30]. We add a many-player dilemma in which the best response to fixed partners can be computed exactly, and real tables of two different models.

Overlap with concurrent work. A concurrent manuscript uses the same game and an overlapping set of models to ask whether models reproduce the risk sensitivity of human players, and suggests that more capable models can respond to risk [3]. It reports a self-play risk baseline related to part of Section 4. The present paper asks how low-cost agents behave against the optimum, against fixed partners and against other models; the tests where paying cannot help and Sections 5 to 7 have no counterpart there.

3 The game and what rational play looks like

3.1 Rules

Six players each start with $e = 40$ units, and the game lasts $R = 10$ rounds. In each round every player privately chooses a contribution $a_i^t \in \{0, 2, 4\}$ to a shared *pool*, and after each round all players see what every seat has paid. Let $x_i = \sum_t a_i^t$ be a player’s total and $X = \sum_i x_i$ the pool. The target is $T = 120$. A player keeps what it did not pay, but if the pool misses the target, a catastrophe destroys all savings with probability p (Table 1 lists the terms we use):

$$u_i = \begin{cases} e - x_i & \text{if } X \geq T, \\ 0 \text{ with probability } p, \text{ else } e - x_i & \text{if } X < T. \end{cases} \quad (1)$$

Proposition 1. *Consider the game in which each player picks a total $x_i \in \{0, 2, \dots, 40\}$, and let $p < 1$. A profile is a Nash equilibrium if and only if $x = 0$, or $X = T$ and $x_i \leq pe$ for every i . A target-reaching equilibrium exists if and only if $p \geq p^* = T/(6e) = \frac{1}{2}$. The welfare-optimal outcome is $x = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, so the best attainable payoff per player is $\text{opt}(p) = \max\{(1-p)e, e - T/6\}$.*

Proof sketch. If $X < T$, any player with $x_i > 0$ gains by paying nothing, so the only equilibrium that misses the target is $x = 0$. If $X > T$, a player can pay 2 less and still reach the target. If $X = T$, the best deviation is to pay nothing, worth $(1-p)e$ against $e - x_i$, which gives $x_i \leq pe$; summing over players gives $p \geq T/(6e)$. Welfare is $6e - X$ when the target is met and $(1-p)(6e - X)$ otherwise, so it peaks at $X = T$ or $X = 0$. The ten-round game, in which strategies can react to the history, has the same totals in pure-strategy Nash

Table 2: Designs. A cell is one condition at one risk level: one model in self-play, one model with one kind of scripted partner, or one pair of models with one split of seats. Each cell holds ten games.

Design	Risk levels p	Games
Self-play grid	0, 0.1, ..., 1	550
In-game questions	0.1, 0.5, 0.9	150
No hint	0.1, 0.5, 0.9	150
Neutral wording	0, 0.1, 0.9	150
Neutral wording, goal stated	0, 0.1, 0.9	150
Two rewordings	0.1, 0.9	200
Temperature zero	0.1, 0.9	100
One model, five scripts (4 kinds)	0.1, 0.3, ..., 0.9	1,000
Two models (10 pairs, 1 to 5 seats)	0.1, 0.5, 0.9	1,500
All		3,950

equilibrium (full proof in the supplementary material); we claim nothing about subgame perfection. $\square$

What we compare play with. Proposition 1 gives the *optimum* for risk-neutral players: keep everything below the pivot and pay the fair share above it. The best response to five fair-share partners switches at the same pivot. The optimum is a benchmark, not a prediction, since paying nothing is an equilibrium at every risk level. Paying is dominated in two situations: at $p = 0$, where both cases of (1) give the same payoff, and once the outcome is settled. Neither test depends on the agent’s attitude to risk or on its beliefs about the others. Payment that continues in these two situations is therefore the sharpest evidence of a default.

Proposition 2. *For two seats i and j in the same game, the difference in their expected payoffs over the lottery is $c(x_j - x_i)$, with $c = 1$ if the target was reached and $c = 1 - p$ otherwise.*

Both seats face the same pool and the same lottery, so only their own payments differ. Hence, for $p < 1$, whoever pays less at a table earns more, whatever the group achieves.

3.2 Models, designs and measures

We test five production models, one per provider, all from the low-cost tier of their family: Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20. We chose this tier because agents that act at scale are likely to run on it. Table 2 lists the designs, and every model plays the same number of games in each of them.

Prompts. In each round a seat receives one prompt with the rules, the full history of every seat’s payments, and a request for one contribution; the supplementary material gives every prompt word for word. The base prompt calls the pool a “climate account”, says that the group “must” reach the target, calls the loss a “disaster” and names the game as a collective-risk social dilemma. It never states the player’s goal, and it hints at the equal split twice. The prompt controls change exactly these parts. *No hint* deletes the two equal-split phrases. *Neutral wording* replaces the normative and naming phrases with plain ones, such as “a shared account”. *Goal stated* keeps the neutral wording and adds an opening that says the

player is in a paid experiment where “the only thing that matters to you is your own final cash payoff”. Two *rewordings* change the phrasing but keep every rule and the hint.

Protocol. The running pool is not printed, so a seat that wants it must add up the history. The prompt is in English, no seat is given a persona, and the temperature is 0.7 unless stated. We read the move from the reply’s final CONTRIBUTION: line. A reply that is cut off at the output limit is asked again with a larger limit rather than guessed, and almost every reply in the data could be read.

Measures. We report *expected* payoffs, the average a seat would earn if the catastrophe lottery were run many times on the same payments, and the share of $\text{opt}(p)$ that is lost. We do not use realised payoffs for a technical reason. The lottery draw in our games depends only on the repetition number, so every cell reuses the same few draws, and by chance all of them fall below one half. Every missed game at $p \geq p^*$ therefore ended in catastrophe, and realised payoffs would punish models that miss more than chance would. Agents never see the draw, so their play is not affected. The unit of analysis is the game, since seats in one game share a history and a lottery. Intervals are 95% percentile bootstraps over games, and tests are permutation tests over games. The supplementary material gives every value behind the statements below.

4 Risk does not change the default

Four of the five models pay about the fair share at every positive risk level (Figure 2a). Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B pay almost exactly the same amount below the pivot as above it, while a risk-neutral player would jump from nothing to the fair share. Risk aversion does not explain this flat line either: preferring a certain fair share to keeping everything at low risk would take strong risk aversion, and no attitude to risk explains paying when there is no risk at all. Gemini 3.5 Flash-Lite is the clearest case. Almost every one of its seats plays the exact split at every risk level (Table 3), so its play is optimal above the pivot only because a fixed rule happens to match the optimum there.

Paying when there is nothing to protect. Four of the five models also pay at $p = 0$, about as much as at any other risk level, although paying there only lowers a player’s own money (Table 3, first column). GPT-5.6 Luna is the exception. It pays nothing at $p = 0$ and its usual share at every positive risk, so it tells no risk apart from some risk but does not respond to how large the risk is.

Three ways to lose. The models lose payoff in three different ways (Table 3, Figure 2c). Gemini 3.5 Flash-Lite, Claude Haiku 4.5 and GPT-5.6 Luna lose mostly at low risk, where they pay for a target that is not worth reaching. Grok 4.20 overpays: its groups reach the target well before the last round, and its seats keep paying into a pool that is already safe. Qwen3-235B fails at the end. It pays 2 round after round, then almost all of its seats pay nothing in the last round (Figure 2b), so its groups almost never reach the target; had each seat simply repeated its previous move, every one of those groups would have succeeded. This is a coordination failure rather than one bad move. Before the last round, the pool of almost every Qwen3-235B table is short of the target by more than one seat can

Table 3: Self-play over the risk grid. Pays at $p=0$: mean seat total where paying is dominated. Exact split: seats paying 2 in all rounds. Target: games reaching it. Payoff: expected payoff per seat; the optimum averages 25.5. Lost: share of $\text{opt}(p)$ lost. Main loss: paying below p^* , paying past 120, or missing the target in the last round.

Model	Pays at $p=0$	Exact split (%)	Target (%)	Payoff	Lost Main (%)
Claude Haiku 4.5	20.9	17	98	18.9	26 below p^*
Gemini 3.5 Flash-Lite	20.0	99.7	100	20.0	21 below p^*
GPT-5.6 Luna	0.0	72	70	19.2	25 below p^*
Qwen3-235B	18.0	4	2	10.9	57 last round
Grok 4.20	37.4	2	100	5.0	80 overshoot

add, so paying nothing is also an equilibrium of that round. Qwen3-235B seats almost always pick it, much as human contributions decline at the end of repeated public goods games [2]. GPT-5.6 Luna shows a milder form of the same failure: some of its groups keep pace until the last round and then miss. Claude Haiku 4.5 shows that the pool can be read from the same prompt, since its seats stop paying in the last round mostly when the target has already been reached. Success rates hide all of this. Gemini 3.5 Flash-Lite and Grok 4.20 both reach the target in every game, yet Grok 4.20 keeps far less of its money, and the panel as a whole loses a large share of the payoff it could have earned.

The default survives rewording, but its failures do not. Deleting the two equal-split phrases lowers no model’s payment (Table 4), so the hint set a ceiling and guided precision but was not the source of the cooperation. Gemini 3.5 Flash-Lite still plays the exact split, which it can rebuild from the target, while GPT-5.6 Luna keeps its average payment but plays the exact split less often. Qwen3-235B and Grok 4.20 pay more without the hint; Qwen3-235B now reaches the target, but it earns less because it overpays. The two rewordings and a temperature of zero leave the payments of the four models other than Grok 4.20 almost unchanged. The failures are less stable than the averages. Under the second rewording Grok 4.20 pays about the fair share, and how often Qwen3-235B drops out in the last round swings widely from one prompt to another. Grok 4.20’s rise with risk on the grid is not stable either: it appears again at temperature zero but mostly disappears in the rerun of the same prompt and under the second rewording, so we do not read it as a response to risk.

Stating the goal, not removing the norms, stops most payment at zero risk. The default does not come from the normative words. With neutral wording, most models still pay at $p = 0$, and at low and high risk four models pay on average what they paid before, although GPT-5.6 Luna plays the exact split less often, while Grok 4.20 pays even more (Table 4). Adding the stated goal is what changes play at zero risk: Claude Haiku 4.5 and Gemini 3.5 Flash-Lite then pay almost nothing there, and Qwen3-235B pays far less than before. The goal still does not make most models weigh the risk. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite pay the fair share already at low risk, so they tell no risk apart from some risk but do not grade it. Only GPT-5.6 Luna moves from paying little at low risk to paying

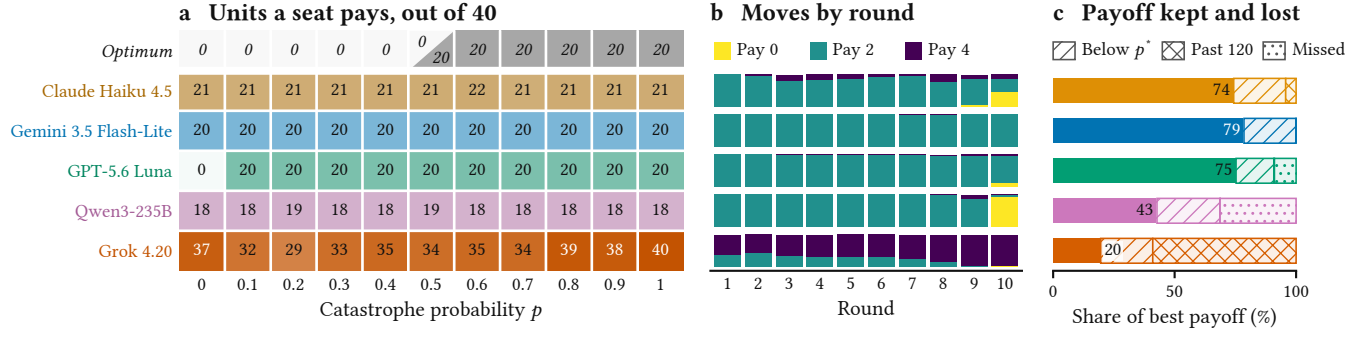


Figure 2: Self-play on the risk grid, 110 games per model. (a) Mean units a seat pays out of 40, ten games per cell; the top row is the optimum of Proposition 1, where both 0 and 20 are optimal at $p = 0.5$. (b) Share of seats paying 0, 2 or 4 in each round, over the games with $p > 0$. (c) Expected payoff kept as a share of $\text{opt}(p)$, and the loss split into paying toward the target below p^* , paying past 120, and missing the target at or above p^* .

Table 4: Mean seat total under prompt and decoding changes. Top: pooled over $p = 0.1$ and 0.9 (20 games each). Rerun repeats the base prompt with in-game questions; Para. 1, 2 are the rewordings; Wording is the neutral wording and Goal adds the stated goal. Bold: a shift from Base of at least 2 units with permutation $P < 0.01$. Bottom: single risk levels (10 games each).

Condition	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Base	20.9	20.0	19.9	18.4	35.1
Rerun	21.3	20.0	19.9	18.5	35.6
No hint	21.7	20.0	19.9	34.7	40.0
Para. 1	21.3	20.0	19.9	19.4	31.8
Para. 2	20.9	20.0	19.8	18.0	20.6
Temp. 0	21.2	20.0	20.0	18.3	35.1
Wording	21.0	20.0	19.4	18.7	39.7
Goal	20.0	20.1	13.9	8.2	37.7
Base $p=0$	20.9	20.0	0.0	18.0	37.4
Wording $p=0$	18.0	19.9	0.4	18.1	39.8
Goal $p=0$	0.3	1.4	0.2	4.8	36.7
Goal $p=0.1$	19.8	20.2	7.7	6.4	36.7
Goal $p=0.9$	20.2	20.0	20.1	10.1	38.7

the fair share at high risk. Qwen3-235B pays somewhat more at high risk than at low risk, but less than before at every risk level, and never reaches the target; Grok 4.20 keeps paying far more than needed. The default is therefore what these models do when the prompt leaves their goal unstated. The scripted and mixed designs below use the base prompt and describe this default.

5 Answers track the risk; play does not

A flat response to risk has a cheap explanation: perhaps the agents do not understand the game. We test this in two ways. We ask the playing agent which choice pays more, and we seat it beside partners whose behaviour is fixed, so that the *best response*, the play that earns the most against them, can be computed exactly.

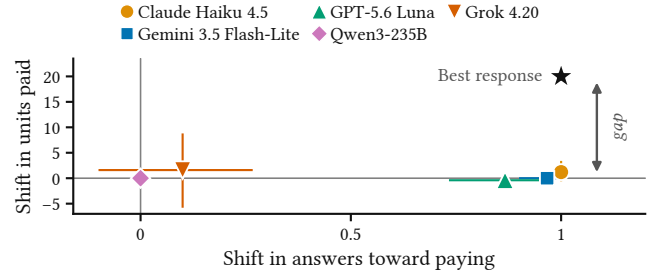


Figure 3: What agents say and do, from $p = 0.1$ to $p = 0.9$ in the same games (ten at each), with 95% intervals. Horizontal: drop in the share of answers saying that paying nothing pays more on average; a correct answerer moves by 1. Vertical: change in what the questioned seat pays. Star: a best responder beside fair-share partners.

5.1 Asking the agent

In the in-game question games we question the first seat at the start, in the middle and at the end of the game, about the state it has just decided on. Each question is a separate call, and its answer never enters the game. Most questions ask about the rules, such as the target or the allowed amounts, and the agents answer all of them correctly. These questions only ask for numbers printed in the prompt, though, so they show that agents read the rules, not that they understand them. We therefore rest on one value question, asked word for word:

Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.

The correct answer is B below p^* , A above it, and equal at p^* . A second value question could be read in two ways, so we report it only in the supplementary material.

Answers move with the risk; play does not. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite answer the value question correctly almost every time, and GPT-5.6 Luna does so most of the time. All three change their answer with the risk (Figure 3, horizontal axis): at low risk they almost always say that keeping pays more, and at high risk they mostly say the opposite. Their payments do not follow (Figure 3, vertical axis): the same seats pay the same amount at low and at high risk. Qwen3-235B and Grok 4.20 show much less of this knowledge. Qwen3-235B always says that keeping pays more, whatever the risk, and Grok 4.20 gives that answer in the first round whatever the risk. The question lays out both strategies side by side, which is easier than finding the comparison inside the game, so the result is a gap between answers and play, not evidence about what an agent computes when it decides.

5.2 Playing against fixed partners

Next we seat one model beside five scripted partners of one kind. They always pay 0, always pay 2, always pay 4, or copy the group, which means paying 2 in the first round and then the rounded average of what the other seats paid in the previous round. The model is not told that its partners are scripted, but their moves are in the history. We found the best response to each kind by trying every possible sequence of the model’s own moves. Beside partners who always pay 0 the target cannot be reached, and beside partners who always pay 4 it is reached without the model, so the best response to both is to pay nothing; once the history settles the outcome, paying is also dominated. Beside fair-share partners the best response is to pay nothing below p^* and exactly the fair share above it, and partners who copy the group never move away from 2 and give the same answer.

No model keeps all its money where that is the best response. In no game beside partners who always pay 0 or always pay 4 does the model pay nothing. Only a minority of all scripted games are exactly optimal, and every one of them is a game in which the optimum is the fair share. The models land on the optimum only where the optimum is the default.

Partners matter; risk does not. We split the variation in what a model pays beside the three fixed-amount partners into parts due to the partner, the risk and their interaction. For Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B, the kind of partner explains most of the variation and the risk almost none, while for an exact best responder in the same games the risk would explain half of it. Beside fair-share partners, where the best response jumps from nothing to the fair share at p^* , no model’s total moves by more than a single payment of 2 between low and high risk. Grok 4.20’s total follows neither the partner nor the risk. Its game is decided by its first move, which it chooses before any history exists: it opens with the largest payment in about half of its games, mostly repeats that opening, and this first move alone explains most of the variation in its total.

Responses to partners take simple forms. The models respond to their partners, but not the way a best responder would (Figure 4, Table 5). Gemini 3.5 Flash-Lite and Claude Haiku 4.5 pay their usual rate and then stop once the history shows that the outcome is settled. GPT-5.6 Luna pays little even before that point, above

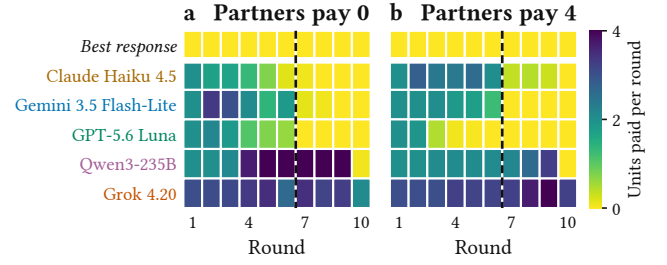


Figure 4: One model beside five scripted partners, pooled over five risk levels (50 games per row). A cell is the mean payment in one round; the top row is the best response, 0 in every round. The dashed rule follows round 6, after which the outcome is settled in most games.

Table 5: Units a model pays per game before and after the outcome is settled, beside five partners who always pay 0 or always pay 4 (50 games per entry). Given these partners, every unit paid is wasted; units paid after settlement are wasted even for an agent that does not know its partners are scripted.

Model	Always 0		Always 4	
	Before	After	Before	After
Claude Haiku 4.5	7.7	0.1	14.2	1.1
Gemini 3.5 Flash-Lite	13.6	0.2	10.9	0.0
GPT-5.6 Luna	8.5	0.0	4.5	0.0
Qwen3-235B	17.8	12.1	12.3	8.1
Grok 4.20	19.6	11.1	16.7	16.2

all beside partners who always pay 4, which comes close to a best response while it learns who its partners are. Qwen3-235B pays more, not less, beside partners who pay nothing. Qwen3-235B and Grok 4.20 keep paying large amounts after the outcome is settled, which no belief about the partners can justify. Qwen3-235B’s last-round rule also costs its partners. Beside five fair-share partners it pays 2 for nine rounds and then nothing, so the table misses the target in almost every game, each time by that single missing payment. Gemini 3.5 Flash-Lite in the same seat reaches the target every time.

6 Defaults in mixed groups

Self-play and scripted partners cannot show what happens when agents with different defaults share a table. We therefore seat every pair of models together, with one to five seats for the first model, at low, moderate and high risk. The seats of each model form one block whose position is fixed within a cell and alternates across cells, so every model holds every seat equally often overall.

At a shared table every model earns more than Grok 4.20, while the payoff gaps among the other four models are small (Figure 5, above the diagonal). This follows from Proposition 2: Grok 4.20 pays the most, and at one table the larger payer earns less. Group success depends on the pair (Figure 5, below the diagonal). Tables that include Grok 4.20 reach the target in the great majority of

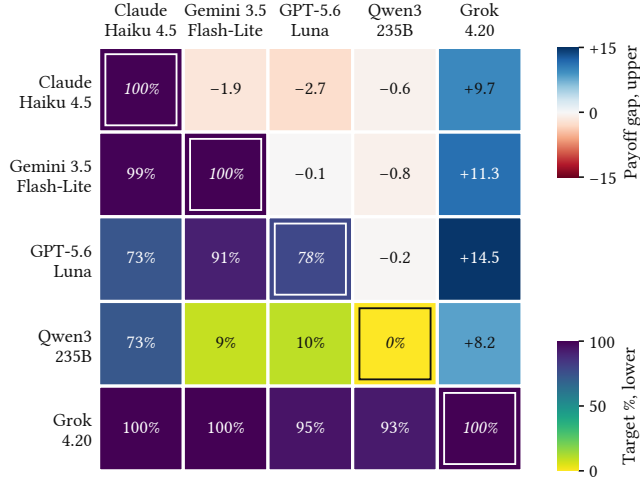


Figure 5: Two models at one table, 150 games per pair, pooled over seat splits and risk. Above the diagonal: expected payoff of the row model’s seats minus the column model’s, in the same games. On and below it: share of games reaching the target; boxed cells are self-play (60 games).

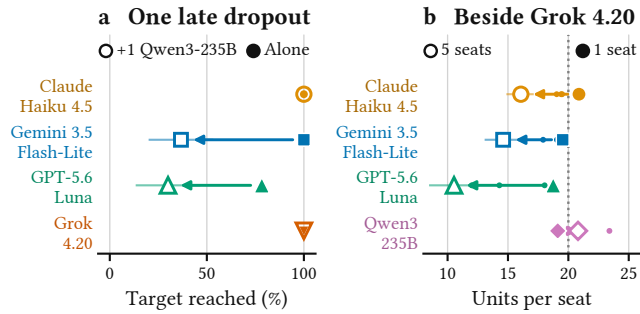


Figure 6: Two models at one table, pooled over p . Arrows run from the filled marker to the open one, which carries a 95% interval. (a) Target reached by six seats of one model (60 games) and by five seats plus one Qwen3-235B seat (30). (b) What a seat pays beside one and five Grok 4.20 seats (30 games each; dots: two to four); dotted: the fair share.

games, while tables that mix Qwen3-235B with Gemini 3.5 Flash-Lite or GPT-5.6 Luna rarely do.

An exact fair share has no slack. One partner that stops paying in the last round is enough to break a table of exact fair-share players (Figure 6a). When the pool is one round of fair shares short of the target before the last round, Qwen3-235B almost always pays nothing, while Gemini 3.5 Flash-Lite almost never does. Five Gemini 3.5 Flash-Lite seats with one Qwen3-235B seat therefore rarely succeed, although Gemini 3.5 Flash-Lite alone never misses, and the same happens to GPT-5.6 Luna. Tables of Claude Haiku 4.5 or Grok 4.20, which pay a little more than needed, absorb the same partner. Almost every missed table with a Qwen3-235B seat would have reached the target had that seat paid 2 in the last round.

Slack also works the other way: one Grok 4.20 seat lets five Qwen3-235B seats reach the target in most games, and Grok 4.20 pays for the rescue. Since Qwen3-235B’s last-round rule depends on the wording, this measures what one late dropout does to a group, whichever model plays that role.

Generous partners are under-matched. Agents pay less, not more, when generous partners fill most of the table (Figure 6b). Next to five Grok 4.20 seats, GPT-5.6 Luna pays about half of what it pays next to other partners, and Gemini 3.5 Flash-Lite and Claude Haiku 4.5 also fall below the fair share. As with scripted partners, Gemini 3.5 Flash-Lite and Claude Haiku 4.5 stop once the pool reaches the target, while GPT-5.6 Luna pays less from the start, so the generous partner carries the group. Qwen3-235B pays slightly more beside Grok 4.20. These responses fit an agent that watches the pool, which our design cannot separate from an agent that tracks who paid.

7 Which agent wins under selection

Systems that host many agents keep some and drop others: an operator picks the agent that earned most, or users copy the one that did well. What matters is whom an agent is compared with, and we show that this choice decides which of our five models such a process favours. Following empirical game-theoretic analysis [29, 31], we treat each model as a strategy. We write $\pi_A(k)$ for the mean expected payoff of an A seat at a table with k seats of A and $6 - k$ seats of another model B , estimated from the mixed tables for $1 \leq k \leq 5$ and from self-play for $k = 6$.

Selection process. A population of N agents holds i agents of type A and $N - i$ of type B , and tables of six are drawn at random. The fitness of each type is its payoff averaged over the tables it can land in:

$$f_A(i) = \sum_{k=1}^6 \mathcal{H}(k-1; N-1, i-1) \pi_A(k), \quad (2)$$

$$f_B(i) = \sum_{k=0}^5 \mathcal{H}(k; N-1, i) \pi_B(6-k),$$

where $\mathcal{H}(k; N-1, m)$ is the hypergeometric chance that k of the five other seats come from the m other agents of type A . An agent copies a random other agent with probability $1/(1 + e^{-\beta \Delta f})$, where Δf is their fitness gap and the selection intensity β is 1 per payoff unit. α -Rank [18] assumes that a new model enters rarely, so the population almost always consists of one model. It computes the chance that a single newcomer takes over, for every pair of models, and turns these chances into a Markov chain over the five one-model populations. The share of time the chain spends on a model is the weight selection gives it, its *stationary mass*. The population size sets whom an agent is compared with. With $N = 6$ the whole population is one table, so an agent is compared only with its tablemates. With $N = 30$, fitness is averaged over random tables, so an agent is compared mostly with agents who sat elsewhere.

Scoring against tablemates rewards the smallest payer. Within one table, Proposition 2 fixes the ranking before any data are seen: the seat that pays less earns more. What the data add is who pays least in company, which is Qwen3-235B, and what this selection costs the group. Selection gives Qwen3-235B the largest weight at every risk level (Figure 7, left column). The selected population then reaches the target far less often than a model picked at random, and stronger selection pushes group success close to zero. This is

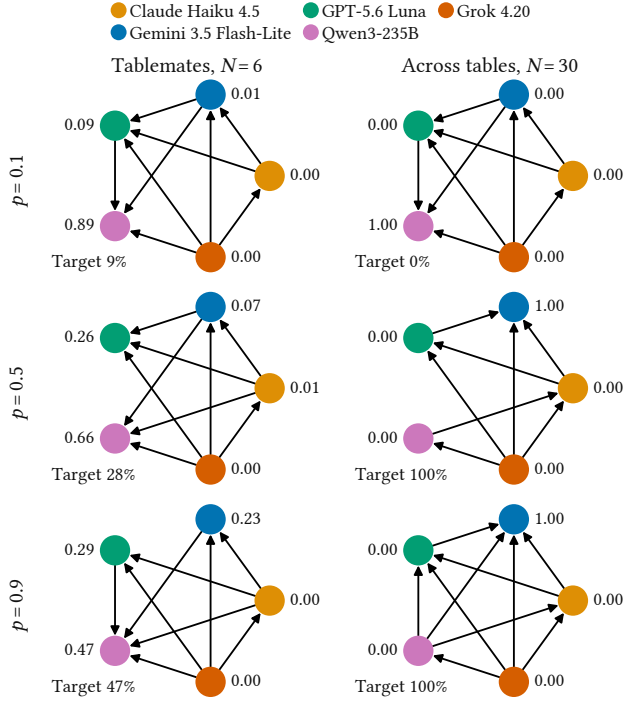


Figure 7: Which model selection favours (α -Rank, $\beta = 1$), computed and drawn with EGTtools [8]. A node is a one-model population, with its stationary mass beside it. An arrow from A to B means that one B newcomer takes over an A population with more than the neutral probability $1/N$. Left: $N = 6$, so agents are compared with their tablemates. Right: $N = 30$, so mostly with agents at other tables. Bottom left of each graph: the self-play target rate, weighted by mass.

the known result that selection on payoff relative to a small group favours whoever does better than its neighbours, even at a cost to the group [23], here shown with language-model agents.

Scoring across tables keeps the fair-share player. When agents are compared with agents at other tables, what matters is how well a model’s own tables do. At moderate and high risk, selection then favours Gemini 3.5 Flash-Lite (Figure 7, right column), whose population always reaches the target and earns the optimum. It wins because it is rigid: a newcomer that pays less pushes its table below the target, and one that pays more loses money. At low risk both rules favour Qwen3-235B, which still earns far less than the optimum, since every model pays at low risk. These winners do not change with stronger selection, with larger populations, or when self-play payoffs come from the risk grid alone, and each stays on top in the large majority of bootstrap samples. Under weak selection no model gains much weight within a table, so the cost of tablemate scoring needs selection that is at least moderately strong. The same agents, scored in two reasonable ways, lead to opposite outcomes for the group, in line with theory on selection between groups [28], and tablemate scoring is one simple route to the selfish outcomes that imitation produced for model-written strategies [32].

8 Discussion

Under the base prompt no model grades its payments by the size of the risk, and four models keep paying where paying helps no one, so generosity alone cannot explain their play. The default appears when the goal is left unstated, not because of normative wording, and all five models adjust to partners only in simple ways.

Possible causes. Our data record choices, not reasoning, so we offer hypotheses. First, the equal split may be a focal point, a choice everyone expects everyone else to make; models rebuild it without the hint, and an even split can likewise reflect a focal point rather than a preference [6]. Second, a question may call up computation while a move request calls up a default, which fits reports that models can hold a correct belief or plan and act against it [7, 24]. Third, post-training may favour doing one’s share when nothing says otherwise, which fits the finding that models are more cooperative than people [16] and our finding that naming the goal removes most payment at zero risk. Stronger reasoning is not an obvious fix, since reasoning models free-ride more in public goods games [11].

Consequences for design. State the agent’s goal. An agent that is told only the rules falls back on a default, and one sentence naming its goal removed the payment at zero risk that neutral wording left in place. Evaluations should report what the agent was told to want. *Test against the optimum, not the success rate.* A group can reach its target in every game and still waste most of its money, so a study needs $\text{opt}(p)$, a condition where paying is dominated, such as zero risk, and a check for payment after the outcome is settled. *Test in mixed populations.* Self-play hides how defaults interact: an exact fair-share default is optimal among copies of itself and fragile next to one late dropout, while a generous default absorbs that partner and pays for others. *Choose the selection signal with care.* Any leaderboard, market or imitation process that scores an agent against its tablemates favours whoever pays least, by Proposition 2, and in our games this lowered group success, while scoring across independent groups favoured the agent whose groups succeed. Each test needs only the game rules, a few scripted policies or a second model, and each caught a failure that self-play success rates did not show.

Threats to validity. The goal sentence is itself an instruction, and we test one phrasing of it, at three risk levels, in self-play only. Two other features of the setup may push models toward paying, and neither is tested here. The decision prompt asks for a final line with the move, so models may deliberate less when deciding than when answering a question. And the game is well known, so the fair share may be recalled rather than derived; the neutral wording removes the game’s name, but the rules still describe it. The same prompts were used for every model, so these features cannot explain why the models differ, but they may raise the common level of payment.

Limitations. We test five low-cost models, one per provider, in English and in one game; more capable models may respond to risk [3], so our claims are about this tier. Each cell holds only ten games, which limits power for the noisiest model, Grok 4.20. Our rewordings test wording, not every prompt design. The welfare losses assume risk-neutral players; the two tests where paying is dominated do not. The scripted partners, mixed tables and selection

analysis use the base prompt only; mixed tables hold two models with fixed seat blocks within a cell, and α -Rank assumes rare entry. For four models the self-play grid was collected separately; a rerun of the same prompt matches it (Table 4). Model text is not reproducible, so we release prompts, configurations, seeds and code, and promise reproducibility of the design rather than of the transcripts.

Ethics. The study involves no human subjects and no personal data. Its findings describe failure modes of deployed systems, and we report them so that designers can test for them before agents are trusted with shared resources.

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